



DexterChem

REVISION SERIES Atomic Structure



NOTE

A dedicated Revision playlist link is in the description box of this chapter.

Containing :

- Revision Video***
- 2022 PYQs***
- 2023 PYQs***

To get Notes + PYQs of this chapter

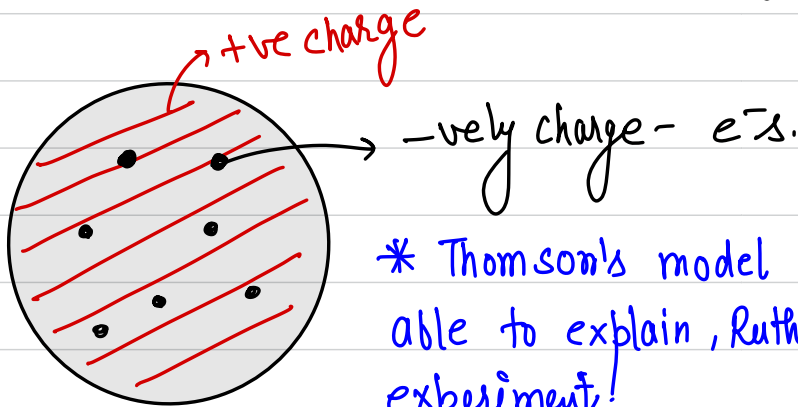
Link in description or go to www.dexterchem.in/courses to find PYQ and Revision Notes.

Structure of Atom

* According to Dalton's theory, atom is the smallest particle that can take part in a chemical reaction.

	<u>Atom</u>		
	Electron	Proton	Neutrons
charge	$-1.6 \times 10^{-19} \text{ C}$	$+1.6 \times 10^{-19} \text{ C}$	0
mass	$9.1 \times 10^{-31} \text{ kg}$	$1.66 \times 10^{-27} \text{ kg}$	$1.66 \times 10^{-27} \text{ kg}$
	J. J. Thomson	Rutherford Goldstone	James Chadwick.

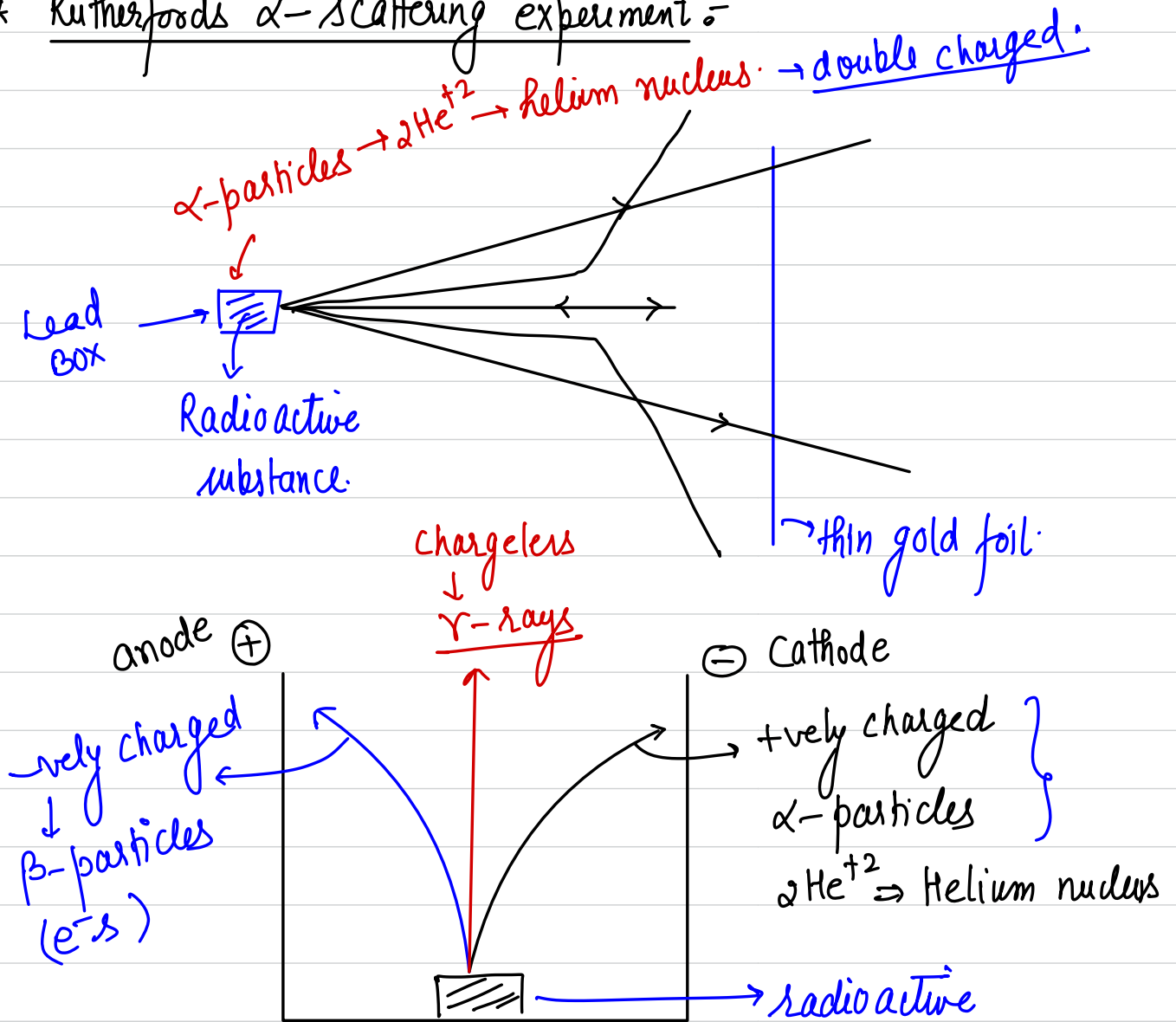
⊛ Thomson's Atomic Model :- (Plum pudding model) :-
(water melon model)



* Thomson's model of atom could not explain, Rutherford's α -scattering experiment!

* It explains the neutrality of the atom.

* Rutherford's α -scattering experiment:-



(*) Observations and main features:-

- ① Most of the α -particles were continued on their straight path that means most of the space inside the atom is empty.
- ② Some of the α -particles when fired closed to the centre got deflected at some angles that means some positive charge is present near the centre.

③ very few particles are thrown back that means all the positive charge is present at the centre & Rutherford called it **Nucleus** and all the mass is also present at the nucleus.

④ The size of the nucleus is very small as compared to the size of atom.

radius of nucleus $r_n = r_0 \times A^{\frac{1}{3}}$

$r_0 \rightarrow$ is a constant.

$A \rightarrow$ Atomic mass of atom

* The radius of nucleus is of the order of 10^{-13} to 10^{-15} m.

* The radius of atom is of the order of 10^{-10} to 10^{-12} m.

$$1 \text{ \AA} = 10^{-10} \text{ m}$$

$$(1 \text{ pm}) \quad 1 \text{ picometer} = 10^{-12} \text{ m}$$

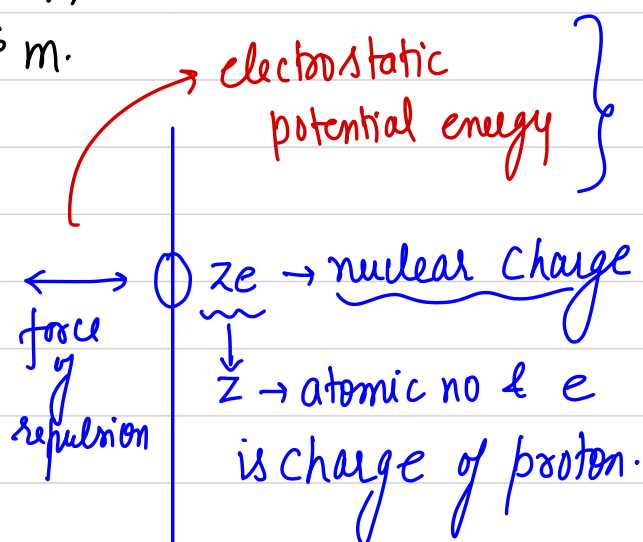
$$(1 \text{ fm}) \quad 1 \text{ femtometer} = 10^{-15} \text{ m}$$

$$1 \text{ nm} = 10^{-9} \text{ m}$$

$$1 \text{ \mu m} = 10^{-6} \text{ m}$$

$$KE = \frac{1}{2}mv^2$$

α -particle



minimum distance of approach:-

when kinetic energy of α -particle = potential energy

$$\frac{1}{2}mv^2 = \frac{k(ze)(2e)}{r}$$

$z \rightarrow$ atomic no
 $e \rightarrow$ charge of proton

$r \rightarrow$ distance of approach
 $2e \rightarrow$ charge of α -particle.

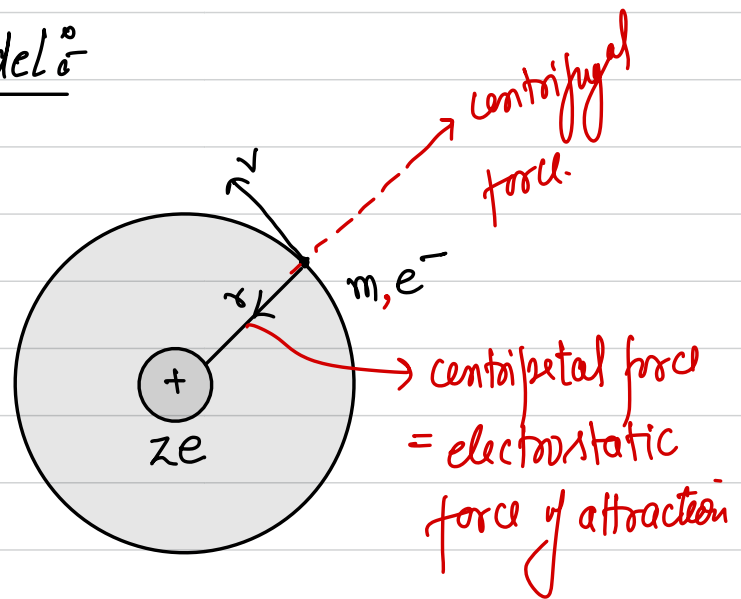
$$r = \frac{k(ze)^2}{\frac{1}{2}mv^2}$$

$$k = \frac{1}{4\pi\epsilon_0}$$

$k \rightarrow$ const. $\epsilon_0 \rightarrow$ epsilon knot
 $\rightarrow$ permittivity of air/space.
 $k = 9 \times 10^9$

⊛ Rutherford's Atomic Model:-

$z \rightarrow$ atomic no
 $e \rightarrow$ charge of proton
 $e^- \rightarrow$ charge of electron
 $m \rightarrow$ mass of electron
 $v \rightarrow$ velocity of electron
 $r \rightarrow$ radius of atom.



$$F_{elec} = \frac{kze^2}{r^2}$$

$$F_{centrifugal} = \frac{mv^2}{r}$$

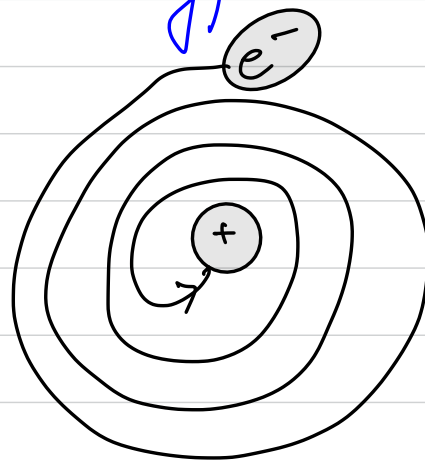
for stability of atom

$$F_{\text{centrifugal}} = F_{\text{elec}}$$

$$\frac{mv^2}{r} = \frac{kZe^2}{r^2}$$

* Maxwell's theory of Electromagnetic radiation

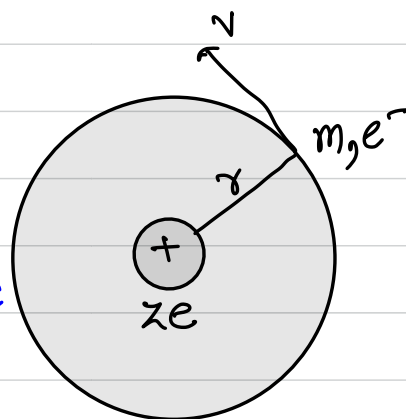
According to this theory, whenever a charge particle moves in an electrostatic field of oppositely charged particle it follows a circular spiral path & radiates energy.



Rutherford was not able to explain this.

Bohr's atomic model :-

Postulates :-



- ① Maxwell's theory of electromagnetic radiation won't be applicable for the circular orbits in which the electron revolves.

In these orbits the e^- will not loose any energy & they are called stationary orbits.

- ② The electron can move only in those orbits whose angular momentum is an integral multiple of $\hbar/2\pi$.

$$\boxed{mvr = n \times \frac{\hbar}{2\pi}}$$

angular momentum

$\hbar \rightarrow$ planck's constant

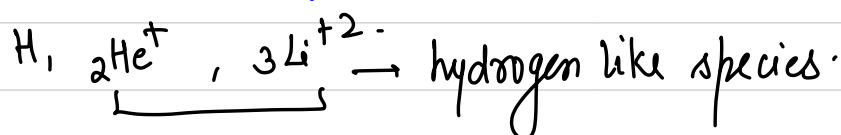
$$\hbar = 6.63 \times 10^{-34} \text{ J-sec.}$$

$m \rightarrow$ mass of e^-
 $v \rightarrow$ velocity of e^-
 $r \rightarrow$ radius of orbit

$n \rightarrow$ orbit no or shell no.

$$n = 1, 2, 3, \dots$$

- ③ This theory is applicable only in case single e^- species or Hydrogen like species.



Bohr's theory:-

$$mvr = \frac{nh}{2\pi} \quad \text{--- (1)}$$

$$\Rightarrow r = \frac{nh}{2\pi} \times \frac{1}{mv}$$

$$\frac{mv^2}{r} = \frac{kze^2}{r^2} \quad \text{--- (2)}$$

$$mv^2 = \frac{kze^2}{r}$$

$$mv^2 = \frac{kze^2}{\frac{nh}{2\pi} \times \frac{1}{mv}}$$

$$mv^2 = \frac{k \times ze^2 \times 2\pi \times mv}{nh}$$

$$v = \frac{k \times ze^2 \times 2\pi}{nh}$$

$$v = \left(\frac{k \times e^2 \times 2\pi}{h} \right) \times \frac{z}{n}$$

~~imp~~ *

$$v = 2.18 \times 10^6 \times \frac{z}{n} \text{ m/s}$$

$$\begin{aligned} k &= 9 \times 10^9 \Rightarrow \\ e &= 1.6 \times 10^{-19} \text{ C} \\ \pi &= 3.14 \\ h &= 6.6 \times 10^{-34} \text{ J-s} \end{aligned}$$

Z - atomic no $n \Rightarrow$ orbit no or shell no
or Principal quantum no.

$\left. \begin{matrix} n=1 \\ Z=1 \end{matrix} \right\} \rightarrow$ hydrogen atom ground $\Rightarrow n=1$.

$$v \propto \frac{z}{n} \quad *$$

$$v = 2.18 \times 10^6 \times \frac{Z}{n} \text{ m/s}$$

$$mv\gamma = \frac{n\hbar}{2\pi}$$

$$\begin{aligned} \gamma &= \frac{n\hbar}{2\pi} \times \frac{1}{mv} \\ &= \left[\left(\frac{\hbar}{2\pi} \times \frac{1}{m} \right) \times \frac{1}{2.18 \times 10^6} \right] \times \frac{n}{Z} \times n \end{aligned}$$

$$* \quad \gamma = 0.529 \times \frac{n^2}{Z} \text{ \AA}$$

$$\gamma \propto \frac{n^2}{Z}$$

for H atom $Z=1$ ✓
ground state $n=1$ ✓

$$\gamma = 0.529 \text{ \AA} = 0.53 \text{ \AA}$$

$\Rightarrow a_0 \Rightarrow$ Bohr's radius

Kinetic Energy $K.E = \frac{1}{2} mv^2$

$$K.E. = \frac{1}{2} m \times \left(2.18 \times 10^6 \times \frac{Z}{n} \right)^2$$

$$K.E. = \frac{1}{2} \times 9.1 \times 10^{-31} \left(2.18 \times 10^6 \times \frac{Z}{n} \right)^2 \text{ J.}$$

$$K.E. = 13.6 \times \frac{Z^2}{n^2} \text{ eV}$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J.}$$

$$\text{Potential energy} = -\frac{kze^2}{r}$$

$$P.E = -27.2 \times \frac{Z^2}{n^2} \text{ eV}$$

$$\begin{aligned} \text{Total Energy} &= K.E + P.E. \\ &= +13.6 \times \frac{Z^2}{n^2} + \left(-27.2 \times \frac{Z^2}{n^2} \right) \end{aligned}$$

$$\boxed{E_n = -13.6 \times \frac{Z^2}{n^2} \text{ eV}}$$

* The negative sign represents that the atom is stable.

$Z=1, n=1$] ground state of H-atom

$$E = -13.6 \text{ eV}$$

The -ve sign represents that the energy of the binding e^- with the nucleus is less than that of free e^- .

* Lesser the energy greater will be the stability.

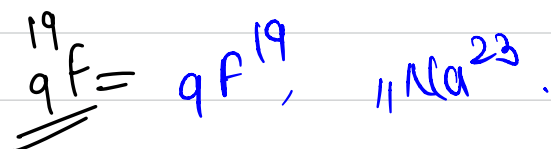
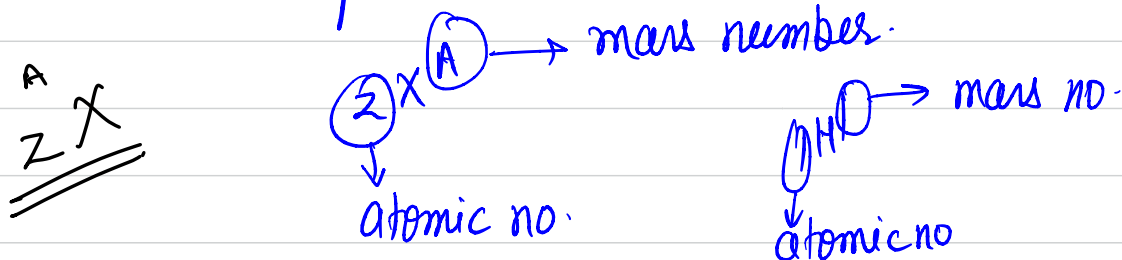
$r_n = 0.529 \times \frac{n^2}{Z} \text{ \AA}$	$r_n \propto \frac{n^2}{Z}$
$v_n = 2.18 \times 10^6 \frac{Z}{n} \text{ m/s}$	$v \propto \frac{Z}{n}$
$E_n = -13.6 \times \frac{Z^2}{n^2} \text{ eV}$	$E_n \propto \frac{Z^2}{n^2}$

Some Basic terms :-

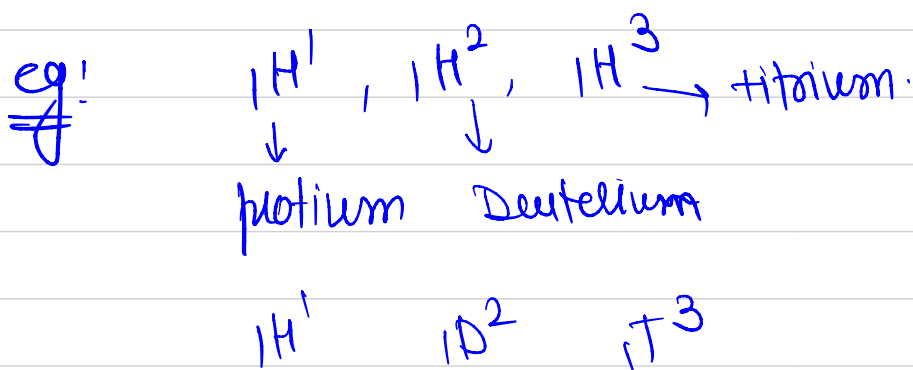
① Atomic number:- (Z) :- No. of protons.

② Mass number $\rightarrow$ (A) :- no of protons + no of neutrons.

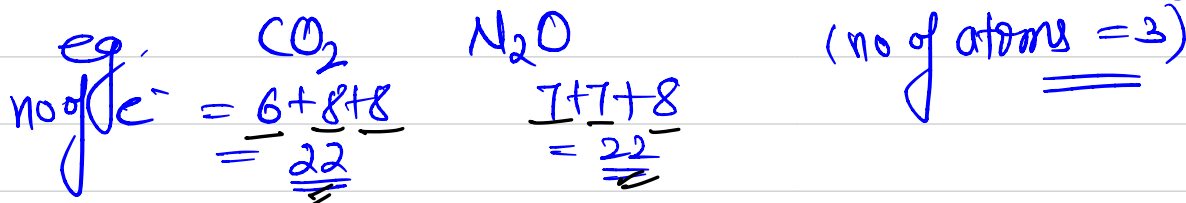
How to represent an element =



③ Isotopes:- the atoms of same element having same atomic no but different mass no.

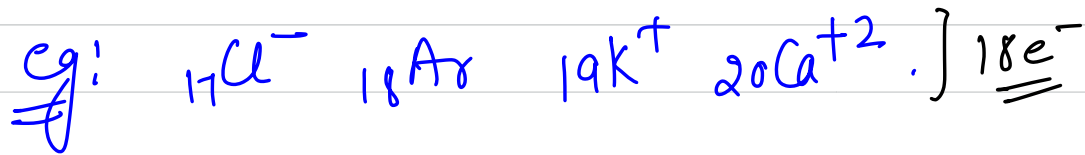


④ Isosters:- Substances which have same number of electrons as well as same no. of atoms in a molecule are called isosters.

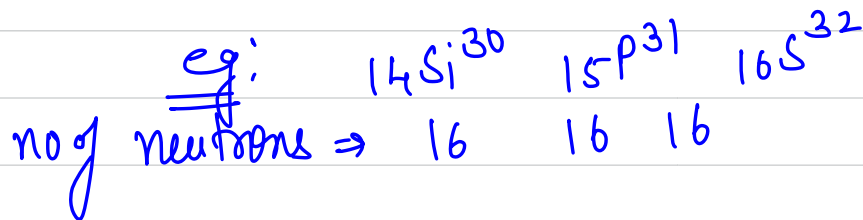


⑤ Isoelectronic :-

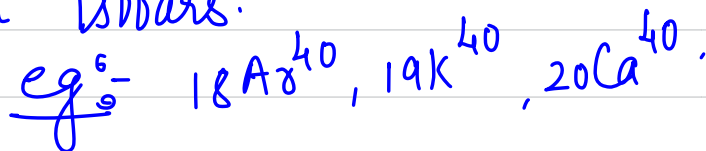
Ion or atom or molecule which have the same number of electrons is called isoelectronic species.



⑥ Isotones :- Elements which contain same no. of neutrons are called isotones.

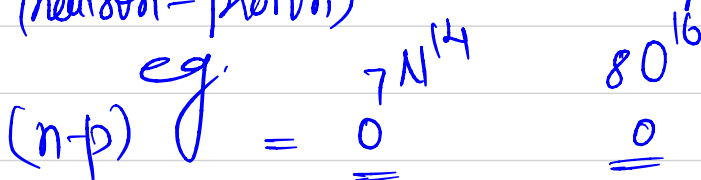


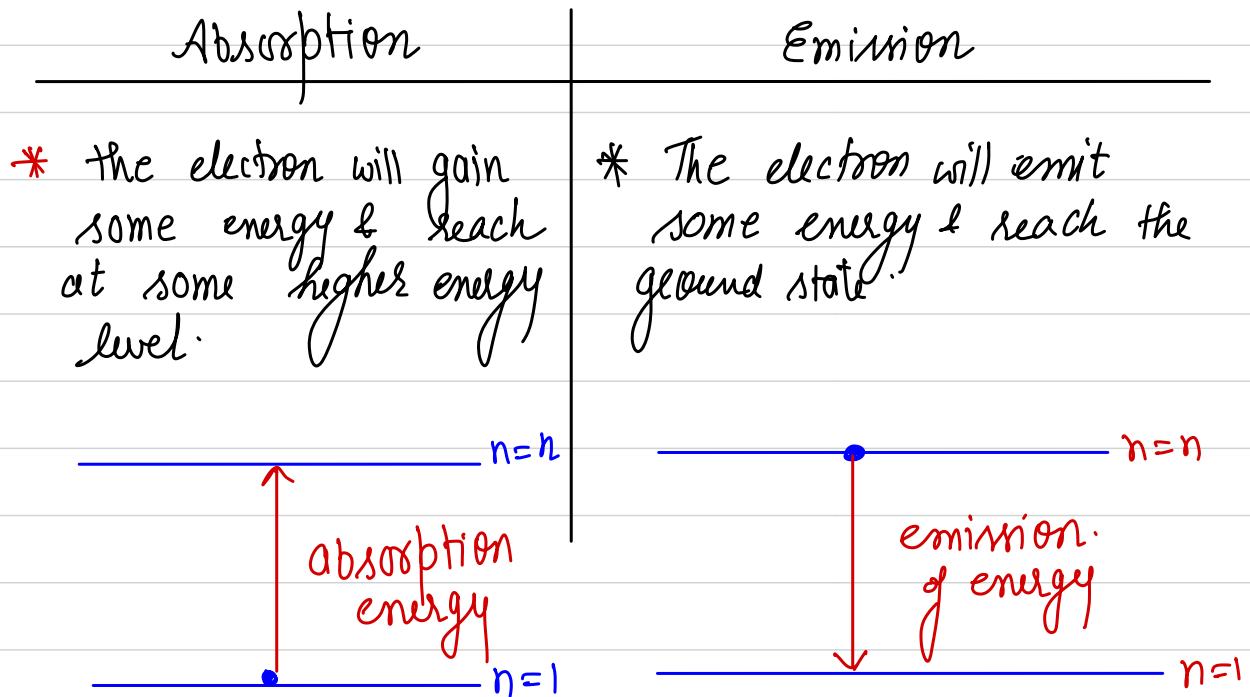
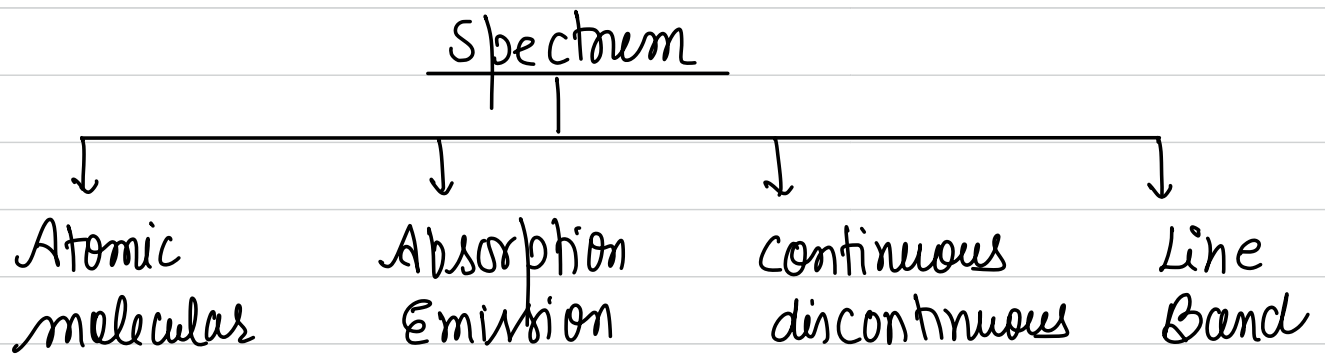
⑦ Isobars :- The two different atoms which have same mass number but different atomic no. are called isobars.



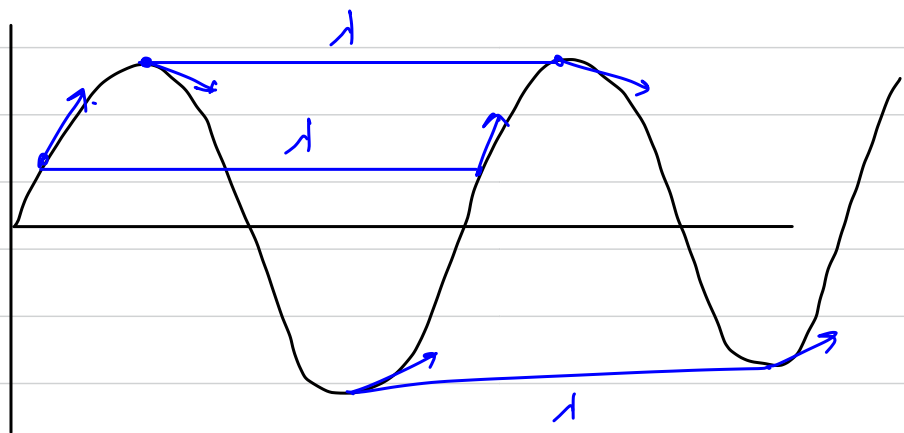
⑧ Isodiaphers :-

The elements which have same value of (n-p) are called isodiaphers.
 (neutron-proton)





Wave length (λ) :-



* wavelength is the distance b/w the two particles on the wave which are in same phase.

Planck's Quantum Theory (Particle nature of light)

Whenever light comes from any source it does not come in continuous manner but it comes in the form of small packets of energy called "photons". The energy associated with every photon is called one quanta.

The energy associated with light is E & the frequency of light used is ν .

$$E \propto \nu$$

$$E = h\nu$$

$h \rightarrow$ planck's constant

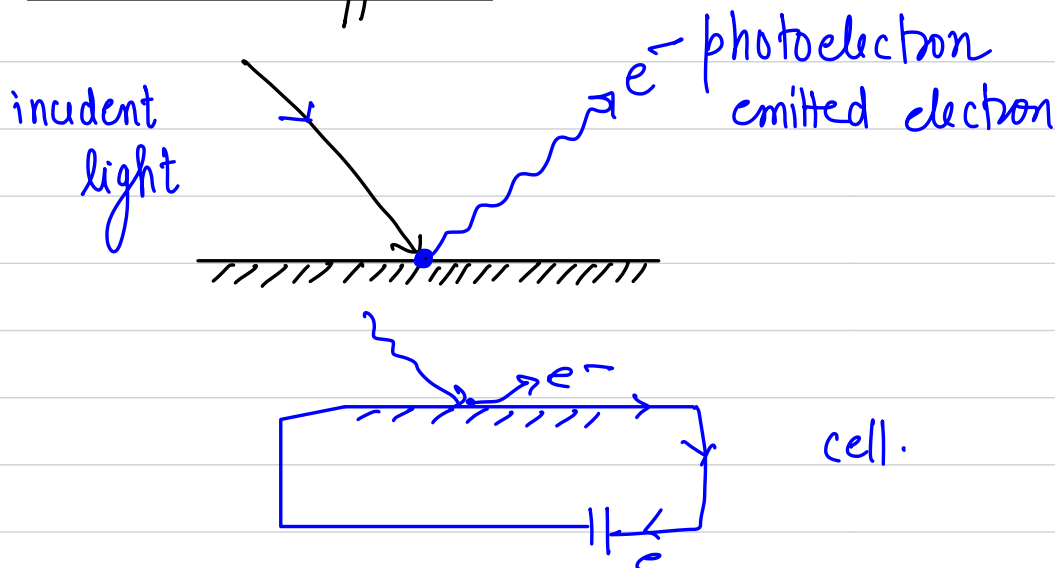
$$E = \frac{hc}{\lambda}$$

$\lambda \rightarrow$ wave length of light used

$$\nu = \frac{c}{\lambda}$$

$c \rightarrow$ speed of light $= 3 \times 10^8 \text{ m/s}$.

Photoelectric effect



The emission of electrons from a metal surface when exposed to a light of radiation of appropriate wavelength is called photoelectric effect.

* Minimum energy required to eject an electron from the metal surface is called 'threshold energy' and the frequency associated with that energy is called "threshold frequency" and is represented by ν_0 .

Let us suppose we choose a light of frequency ν which is greater than ν_0 .

from Law of conservation energy.

Energy given = threshold energy + K.E. of emitted e^-

$$h\nu = h\nu_0 + \underline{\text{K.E.}}$$

$$h\nu = h\nu_0 + \frac{1}{2}mv^2$$

$\phi \Rightarrow$ work function

$$h\nu = h\nu_0 + \frac{1}{2}mv^2$$

$$h(\nu - \nu_0) = \frac{1}{2}mv^2$$

velocity of e^-

$$v = \sqrt{\frac{2h(\nu - \nu_0)}{m}}$$

$$K.E. = \frac{\text{charge} \times \text{voltage}}{e}$$

$$\text{voltage} = \frac{K.E.}{e} = \frac{\frac{1}{2}mv^2}{e}$$

$$\Rightarrow h\nu = h\nu_0 + e(\text{voltage})$$

$$K.E. = eV_{0L}$$

$$\text{voltage} \rightarrow V_{0L} = \frac{h(\nu - \nu_0)}{e}$$

$$h\nu = h\nu_0 + K.E.$$

(*) Important facts about photoelectric effect:-

- ① There is no time lag b/w the incident light and emission of photoelectron.
- ② For the emission of photoelectrons, the frequency of the incident light must be equal to or greater than threshold frequency.
- ③ The rate of emission of photoelectrons from a metal surface is directly proportional to the intensity of the light used.
- ④ The max. K.E. of the photoelectron depends on the frequency of the light used but it is independent of the intensity of incident light.

Graphs related to photoelectric effect:-

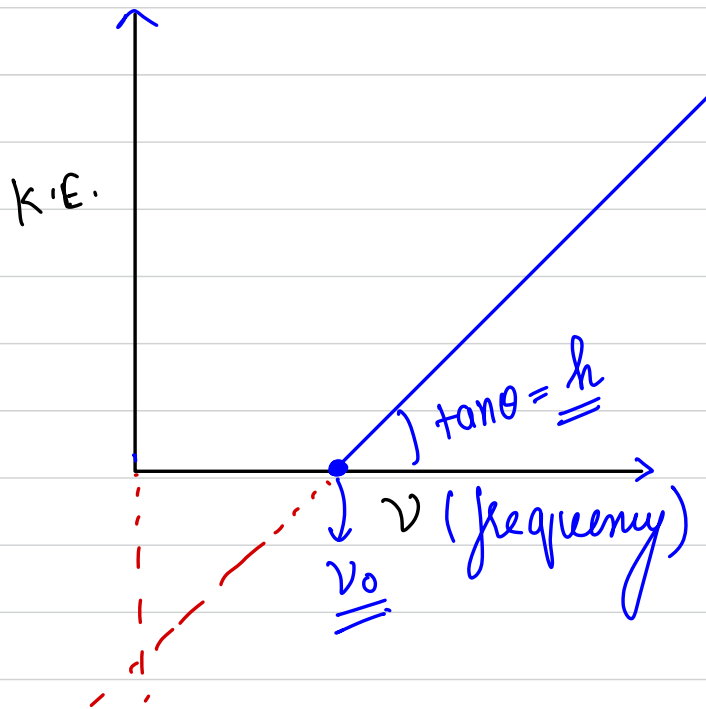
① K.E. vs ν

$$\hbar \nu - \hbar \nu_0 = K.E.$$

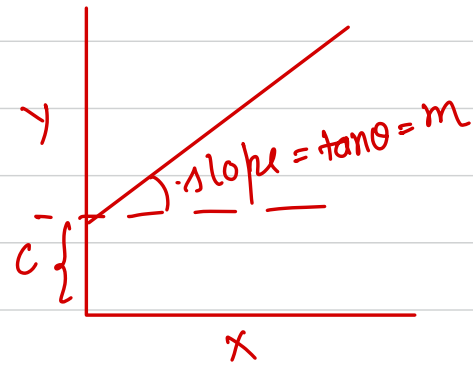
$$m x - c = y$$

$$K.E. = h\nu - h\nu_0$$

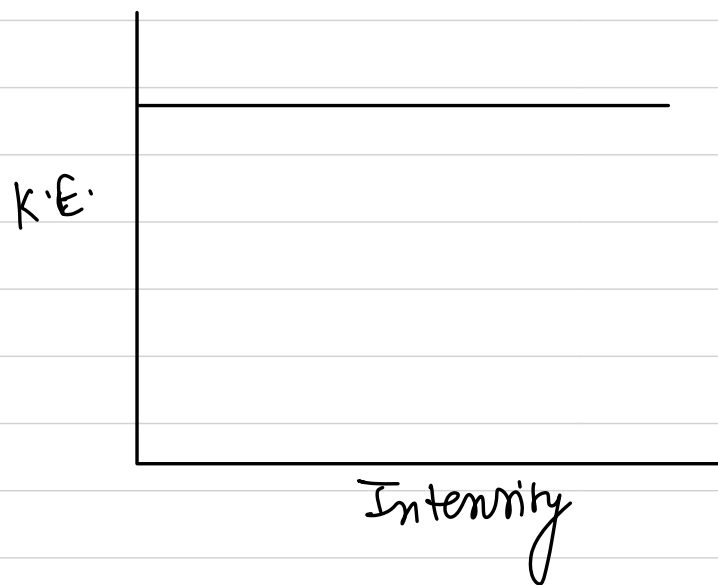
$$y = mx - c$$



$$y = mx + c$$



K.E. vs Intensity:-



Bohr's Theory :-

$$r = 0.529 \times \frac{n^2}{Z} \text{ \AA}$$

$$v = 2.18 \times 10^6 \times \frac{Z}{n} \text{ m/s}$$

$$K.E. = +13.6 \times \frac{Z^2}{n^2} \text{ eV}$$

$$P.E. = -27.2 \times \frac{Z^2}{n^2} \text{ eV}$$

$$E_n = T.E. = -13.6 \times \frac{Z^2}{n^2} \text{ eV}$$

for H atom $Z=1$

$$n=1 \text{ ground state} \leftarrow E_1 = -13.6 \text{ eV}$$

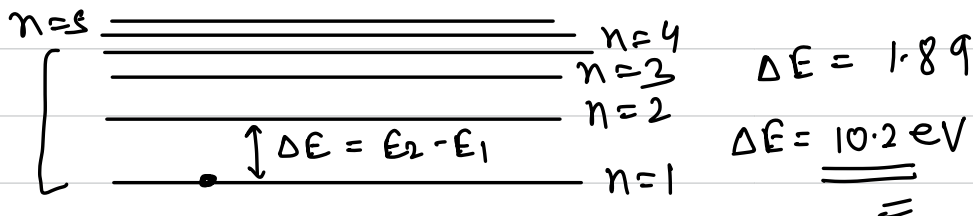
$$n=2 \rightarrow 1^{st} \text{ excited state} \quad E_2 = -3.4 \text{ eV}$$

$$E_3 = -1.51 \text{ eV} \quad n=3 \text{ second excited state}$$

$$E_4 = -0.85 \text{ eV}$$

$\hookrightarrow n=4 \Rightarrow$ third excited state

Quantization of Energy :-

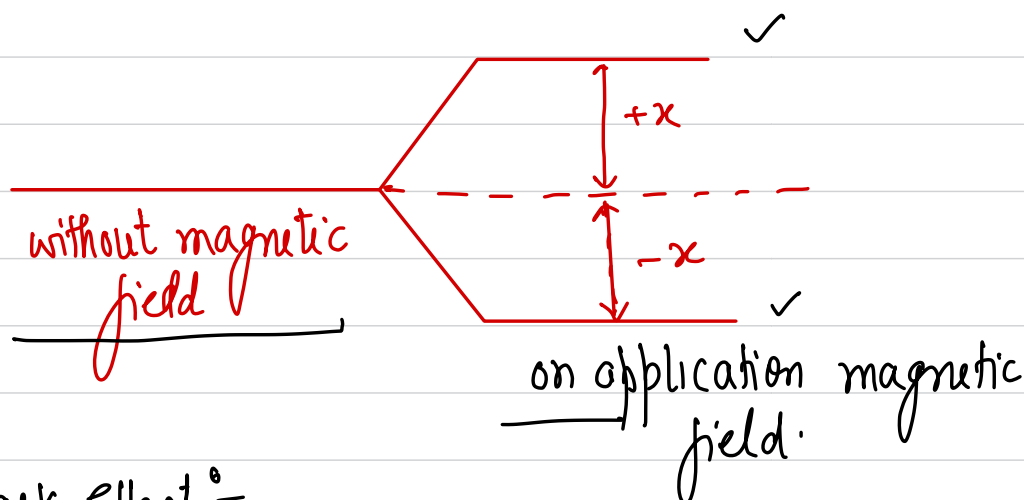


Drawbacks of Bohr's Theory :-

- ① It is applicable only in for single e^- system.
(hydrogen and hydrogen like systems)
ex:- 2He^+ , 3Li^{+2} .
- ② Bohr's theory was not able to explain the fine splitting of spectral lines.
- ③ Bohr's theory was not able to explain 'Zeeman effect' and 'stark effect'.

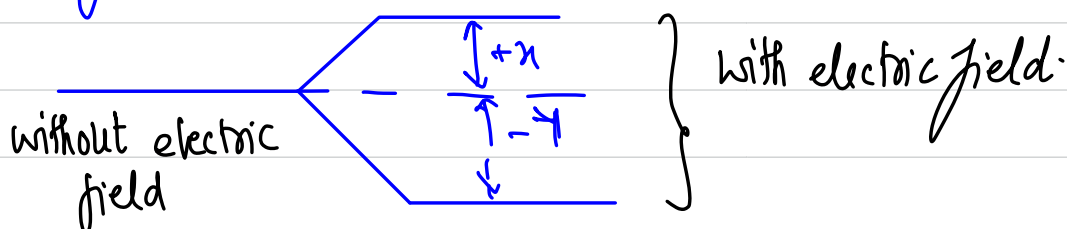
Zeeman Effect :-

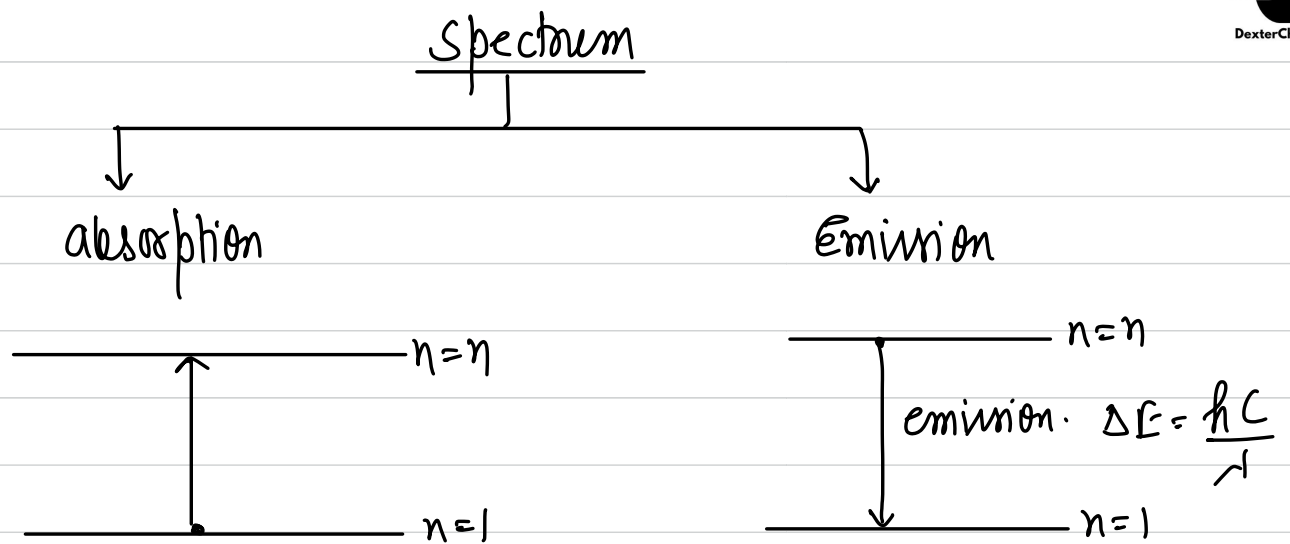
When the spectral lines are exposed to magnetic field they get split.



Stark Effect :-

Splitting of spectral lines in presence of electric field is called stark effect.





$$E_n = -13.6 \times \frac{Z^2}{n^2}$$

$$E_{n_1} = -13.6 \times \frac{Z^2}{n_1^2}$$

$$E_{n_2} = -13.6 \times \frac{Z^2}{n_2^2}$$

$$E_{n_2} - E_{n_1} = +13.6 \times Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\Delta E = 13.6 \times Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\frac{hc}{\lambda} = 13.6 \times Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\frac{1}{\lambda} = \frac{13.6 \times Z^2}{hc} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$= \left(\frac{13.6 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34} \times 3 \times 10^8} \right) \times Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\downarrow$$

$$\bar{\nu} = \frac{1}{\lambda} = 10967800 \, \text{m}^{-1} \times Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

wave number

$$\bar{\nu} = \frac{1}{\lambda} = 1.096 \times 10^7 \times Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\bar{\nu} = \frac{1}{\lambda} = R_H \times Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$R_H \rightarrow$ Rydberg's constant $= 1.096 \times 10^7 \text{ m}^{-1}$

$Z \rightarrow$ atomic no.

$\lambda \rightarrow$ wave length

$\bar{\nu} \Rightarrow$ wave number

$n_1 \Rightarrow$ lower energy level

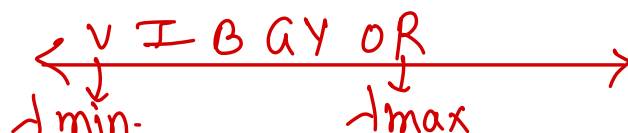
$n_2 =$ higher energy level.

* Balmer Series (Visible Region)

Whenever an electron makes a transition from a higher energy level to $n=2$ then the wavelength of the light is present in visible region.

$$\bar{\nu} = \frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{2^2} - \frac{1}{n_2^2} \right]$$

$$n_2 = 3, 4, 5, \dots$$



* α line of Balmer series $n=3$ to $n=2$
 β line $n=4$ to $n=2$
 γ $n=5$ to $n=2$

② Lyman Series (UV Region)
 e^- jumps from $n_2 = n$ to $n=1$

$$\frac{1}{\lambda} = R_H \times Z^2 \left[\frac{1}{1^2} - \frac{1}{n_2^2} \right]$$

$n_2 = 2, 3, 4, \dots$

③ Paschen Series (Near IR Region)
 $n_2 = n$ $n_1 = 3$

$$\frac{1}{\lambda} = R_H \times Z^2 \left[\frac{1}{3^2} - \frac{1}{n_2^2} \right]$$

$n_2 = 4, 5, 6, 7, \dots$

④ Brackett Series (IR Region)
(Infrared)

$$\frac{1}{\lambda} = R_H \times Z^2 \left[\frac{1}{4^2} - \frac{1}{n_2^2} \right]$$

$n_2 = 5, 6, \dots$

$\lambda_{\max} = ?$

$\lambda_{\min} = ?$

⑤ Pfund Series:- $n_2 = n$ to $n_1 = 5$ (far IR region)

$$\bar{\nu} = \frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{5^2} - \frac{1}{n^2} \right]$$

$$n_2 = 6, 7, 8, \dots$$

$$\lambda_{\min} = ?$$

$$\lambda_{\max} = ?$$

⑥ Humphrey Series:- $n_2 = n$ $n_1 = 6$

$$\bar{\nu} = \frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{6^2} - \frac{1}{n^2} \right]$$

$$n_2 = 7, 8, \dots$$

Calculation of number of spectral lines

When an electron makes a transition from $n = n_2$ to $n = n_1$, then the possible no of spectral lines will be



$$\text{no. of spectral lines} = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

Q: Calculate the no of spectral lines observed when an e^- makes a transition from $n=5$ to $n_1=1$ (ground state).

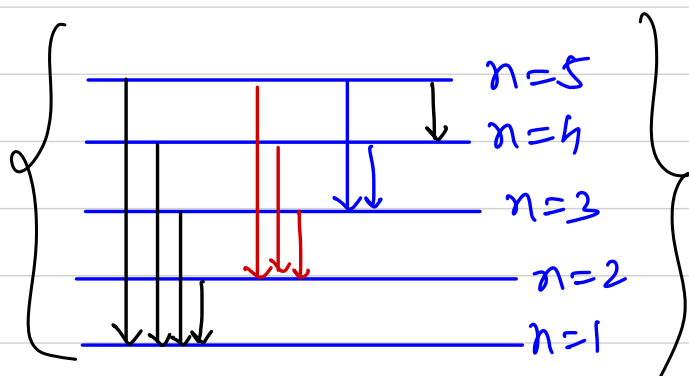
$$n_2 = 5$$

$$n_1 = 1$$

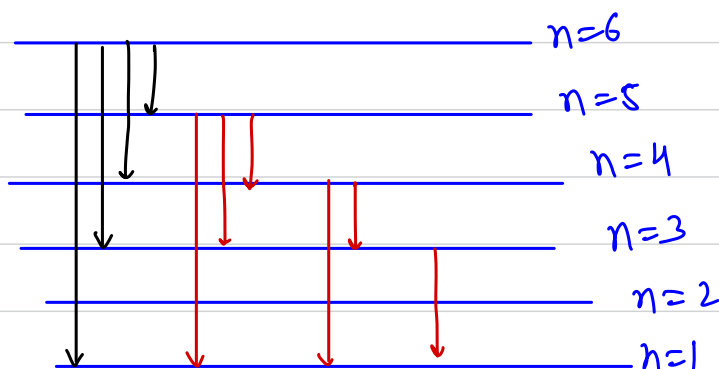
$$\text{no of spectral lines} = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

$$= \frac{(5-1)(5-1+1)}{2}$$

$$= \frac{4 \times 5}{2} = \underline{\underline{10}} \text{ lines.}$$



Q: Calculate the no. of spectral lines obtained from an e^- that makes transitions from $n=6$ to $n=1$ such that no line is observed in visible region.



$$\text{no of lines} = \underline{\underline{10}}$$

De-Broglie's Equation :- (wave particle duality) :-

max planck $\rightarrow$ light has particle as well as wave nature.

De-Broglie $\rightarrow$ particle has wave nature.

According to De-Broglie:-

the wavelength associated with every particle is inversely proportional to the linear momentum of the particle.

$$\lambda \propto \frac{1}{p}$$

$p \rightarrow$ linear momentum
 $p = mv$

$$\lambda \propto \frac{1}{mv}$$

$$\lambda = \frac{h}{mv}$$

$h \rightarrow$ planck's constant.
 $m \rightarrow$ mass of particle
 $v \rightarrow$ velocity of particle.

mass energy relationship

$$E = mc^2 = \frac{hc}{\lambda}$$

$$\lambda = \frac{h}{mc}$$

$$\lambda = \frac{h}{mv}$$

$c \rightarrow$ speed of light
for general particle $c \approx v$

$$\lambda = \frac{h}{mv}$$

$$\lambda = \frac{h}{\sqrt{2mKE}} \quad *$$

$$\lambda = \frac{h}{\sqrt{2meV}}$$

$$\lambda \propto K.E.$$

$$K.E. = \frac{1}{2}mv^2$$

$$2 K.E \times m = \underline{m^2 v^2}$$

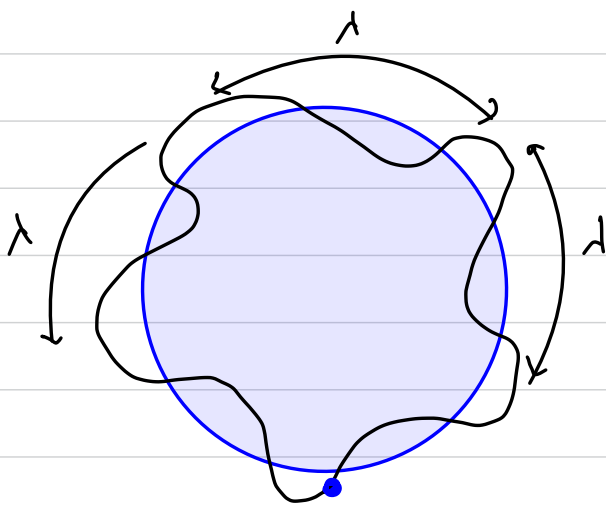
$$mv = \sqrt{2mKE.}$$

$$KE = eV$$

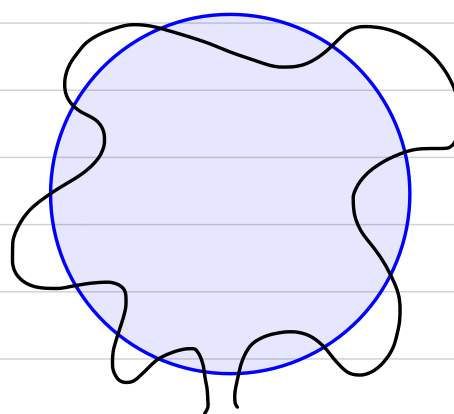
$V \rightarrow$ voltage

Relationship b/w the wavelength and radius of orbit :-

Let us suppose the e^- is moving in circular path (orbit)
Now the electron will behave as a wave.



Electron wave is in phase



Electron wave is out of phase.

The no of waves made by electron = $\frac{\text{circumference of the orbit}}{\text{wavelength}}$.

$$n' = \frac{2\pi r}{\lambda}$$

$$n' = \frac{2\pi r \times mv}{h}$$

$$n' = \frac{2\pi}{h} \times mvr$$

$$n' = \frac{\cancel{2\pi}}{\cancel{h}} \times \frac{n\cancel{h}}{\cancel{2\pi}}$$

$$\boxed{n' = n} \rightarrow \text{orbit no.}$$

↓
no of waves

$$\lambda = \frac{h}{mv}$$

$$mvr = n \frac{h}{2\pi}$$

the no of waves made by e^- in an orbit is equal to its orbit no.

if $n=1$	no of waves = 1
$n=2$	<u> </u> = 2
$n=3$	<u> </u> = 3

$$\text{Bohr's radius} = a_0 = 0.529 \text{ \AA}$$

$$r = a_0 \times \frac{n^2}{Z}$$

the wavelength for n^{th} orbit $\Rightarrow n = \frac{2\pi r}{\lambda}$

$$\lambda = \frac{2\pi r}{n}$$

$$\lambda = \frac{2\pi}{n} \times \left(a_0 \times \frac{n^2}{Z} \right)$$

$$\lambda = 2\pi a_0 \times \frac{n}{Z} \quad \text{for } \underline{n=1} \text{ \& } \underline{Z=1}$$

$$\lambda = 2\pi a_0$$

for $Z=1$

$$\lambda = 2\pi a_0 \times n \neq$$

Q: Calculate the wavelength of the 2nd excited state of H-atom if the wavelength of the ground state is $2\pi a_0$.

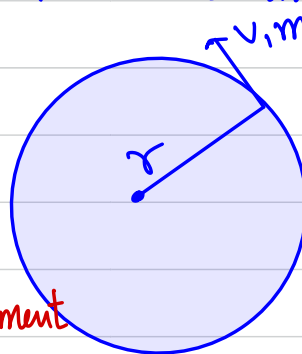
2nd excited state means $\Rightarrow n=3$.

$$\lambda = 2\pi a_0 \times 3 = \underline{\underline{6\pi a_0}}$$

Heisenberg's Uncertainty Principle :-

According to this principle
It is impossible to tell the exact position and exact momentum of a micro particle (e^-) simultaneously.

$$\Delta x \times \Delta p \geq \frac{h}{4\pi}$$



$\Delta x \rightarrow$ uncertainty in the measurement of position

$\Delta p \rightarrow$ uncertainty in the measurement of momentum

$$\frac{h}{4\pi} = \frac{\hbar}{2}$$

$$\hbar = \frac{h}{2\pi}$$

$$\hbar = h \text{ cut}$$

$$\Delta x \times \Delta p \geq \frac{\hbar}{2}$$

(HW)

Q: If the uncertainty in the measurement of energy is ΔE & the uncertainty in the measurement of time is Δt . then establish a relation b/w ΔE & Δt by using Heisenberg's uncertainty Principle.

$$\Delta x \times \Delta p \geq \frac{h}{4\pi}$$

$$F = \frac{\Delta p}{\Delta t}$$

$$\Delta x \times \Delta p \times \Delta t \geq \frac{h}{4\pi}$$

$$\Delta F \times \Delta x = \Delta W$$

$$= \Delta E$$

$$\Delta x \times \Delta F \times \Delta t \geq \frac{h}{4\pi}$$

$$\Delta E \times \Delta t \geq \frac{h}{4\pi}$$

Black Body Radiations :-

Nature of black body radiation can not be explained by classical mechanics.

Max planck in 1890 explained the phenomenon of black body radiation.

- * When solids are heated they emit radiation over a wide range of wavelengths.

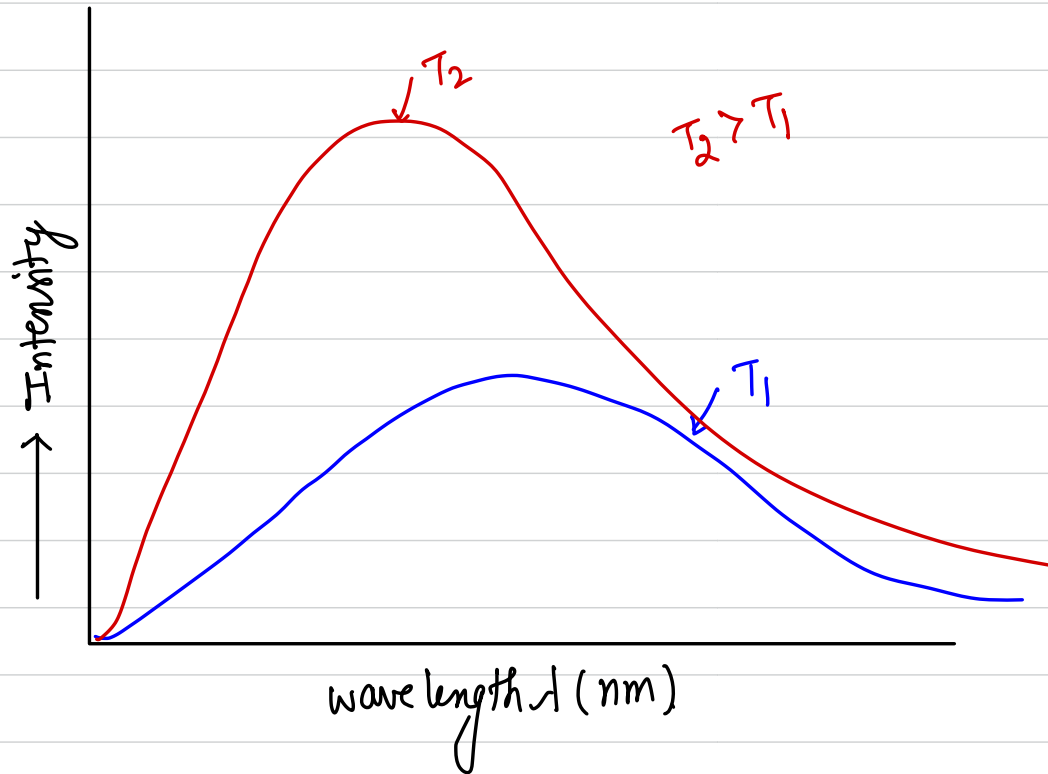
For example:- Iron rod $\xrightarrow{\Delta}$ dull red $\xrightarrow{\Delta}$ more red
 $\downarrow$ $\downarrow$
wavelength $\uparrow$ $\downarrow$
 $\nu = \frac{c}{\lambda}$ $\nu \downarrow$ Blue $\xleftarrow{\Delta}$ white
 $\nu \propto \frac{1}{\lambda}$ at very high temp

- * In terms of frequency, it means that the frequency of emitted radiation goes from a lower frequency to a higher frequency.

- * The ideal black body, which emits and absorbs radiations of all frequencies, is called a black body and the radiations emitted by such a body is called black body radiations.

- * The intensity vs frequency curve of the radiation from a black body depends only on its temperature.

- * At a given temperature, intensity of radiation emitted increases with increase of wavelength, reaches a max. value at a given wavelength and then starts decreasing with increasing wavelength.



Quantum Mechanical Model of atom:-

↓
Schrodinger → 1926 → wave Equation

It is considering the wave nature of particle (e^-).

Eigen value
Eigen function
Relationship.

$$H\psi = E\psi$$

$$\psi \rightarrow \underline{\underline{\psi}}$$

Time independent schrodinger wave equation.

$\psi \rightarrow$ wave function

$H \rightarrow$ Hamiltonian operator.

$E \rightarrow$ Energy value

$$\frac{d}{dx} \underline{e^x} = \underline{1 \cdot e^x} \rightarrow \text{eigen value eigen function relation}$$

$$\frac{d}{dx} \sin x = 1 \cdot \cos x \rightarrow \text{this is not.}$$

$$\frac{d^2}{dx^2} \underline{\sin x} = \underline{-1 \cdot \sin x} \rightarrow \text{eigen value eigen function relation.}$$

$\psi \rightarrow$ It is called the wave function. It is a mathematical function whose value depends upon (x, y, z)

$$\psi(x, y, z)$$

* ψ does not have any physical significance.

Imp * $\psi^2 \rightarrow$ wave function square $\rightarrow$ It gives us the probability density of finding an e^- in a small volume dV .

$$\text{Probability density} = \frac{\text{Probability}}{\text{volume}}$$

$$\text{Probability} = \text{Probability density} \times \text{volume}.$$

$$dP = \psi^2 dV$$

$$H\psi = E\psi$$

del $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi = \frac{8\pi^2 m}{h^2} \underbrace{(V-E)}_{\text{K.E.}} \psi$

$V \rightarrow$ potential energy

$E \rightarrow$ Total energy

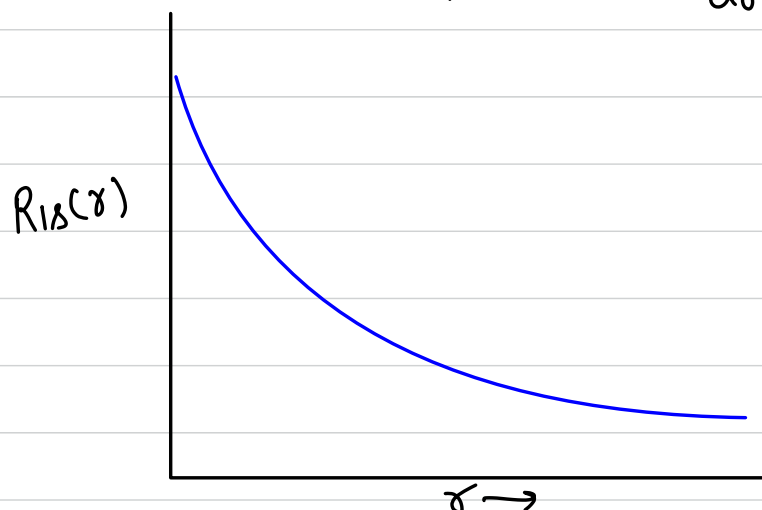
$m \rightarrow$ mass of e^-

$h \rightarrow$ planck's constant.

$$R_{1,0}(r) = R_{1s}(r) = 2 \left(\frac{Z}{a_0} \right)^{3/2} e^{-Zr/a_0}$$

for H atom
 $Z=1$

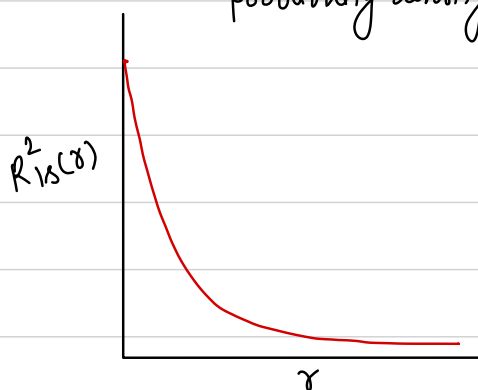
$Z \rightarrow$ nuclear charge
 $a_0 \rightarrow$ Bohr's radius



Wave function does not have any physical significance.
But its square gives us the probability density.

$$R_{1s}^2(r) = 4 \left(\frac{Z}{a_0} \right)^3 e^{-2Zr/a_0}$$

probability density vs r $\rightarrow Z=1$

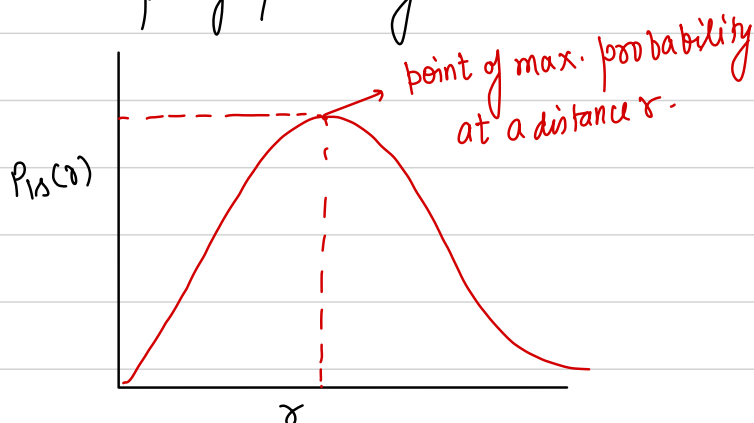


Probability = Probability density $\times$ Volume

$$dP(r) = R_{1s}^2(r) \times 4\pi r^2 dr$$

$$dP_{1s}(r) = 4 \times \left(\frac{Z}{a_0} \right)^3 e^{-2Zr/a_0} \times 4\pi r^2 dr$$

Graph of probability vs r .



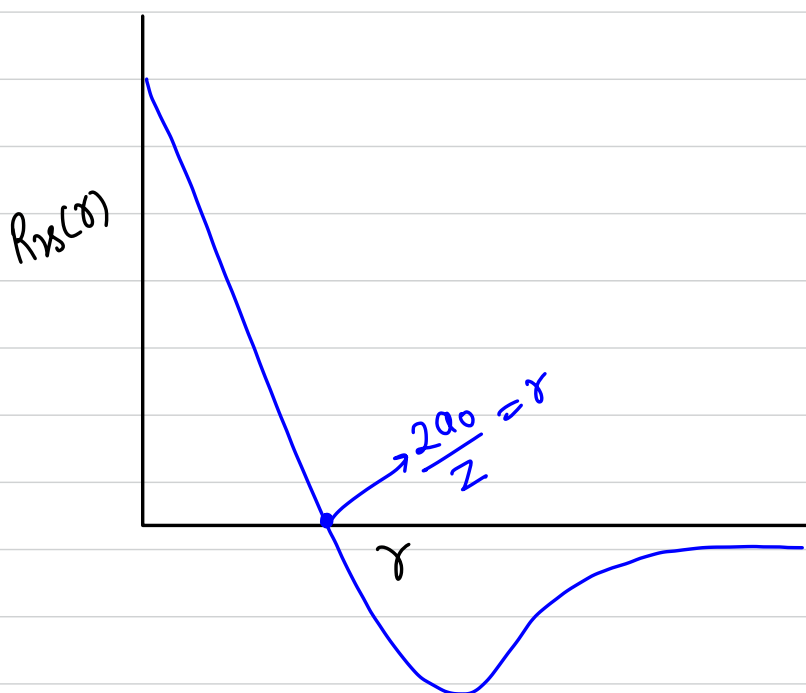
for 2s-orbital:-

$$R_{2s}(r) = R_{2,0}(r) = 2 \left(\frac{Z}{a_0} \right)^{3/2} \left(1 - \frac{Zr}{2a_0} \right) e^{-Zr/2a_0}$$

$R_{2s}(r)$ vs r

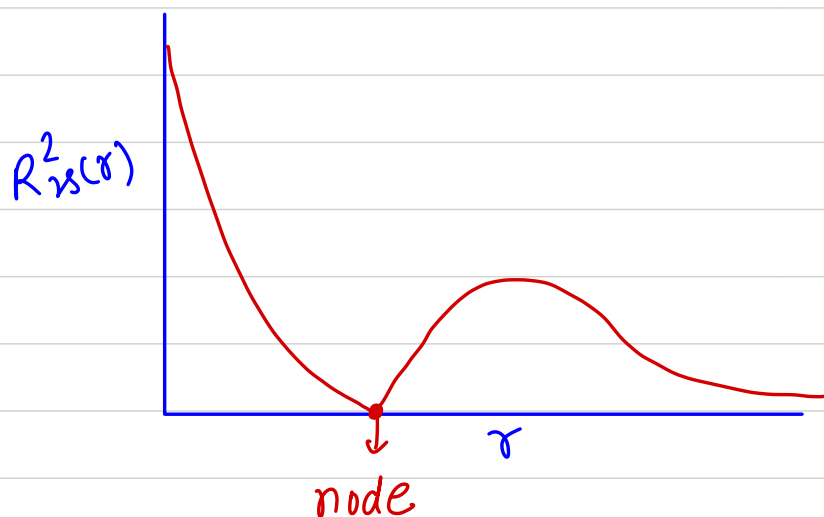
$$1 - \frac{Zr}{2a_0} = 0$$

$$r = \frac{2a_0}{Z}$$



Node:- Node is the space where the probability of finding an electron is zero.

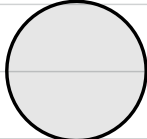
$$* \quad R_{2s}^2(r) = 4 \left(\frac{Z}{a_0} \right)^3 \left(1 - \frac{Zr}{2a_0} \right)^2 e^{-Zr/a_0}$$



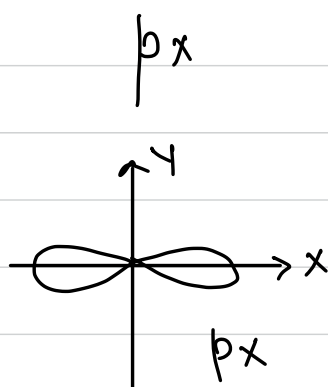
$P_{2s}(r)$ vs r



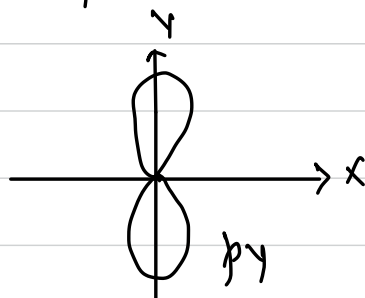
Shapes of Subshell:-

s-subshell $\rightarrow$  spherical

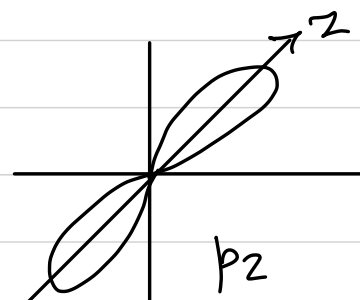
p-subshell $\rightarrow$  dumbbell.
 node



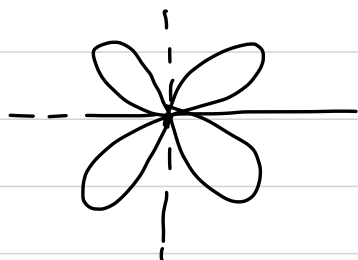
p_y



p_z



d-subshell $\rightarrow$ double dumbbell

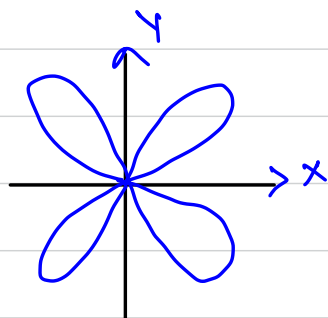


d-subshell has

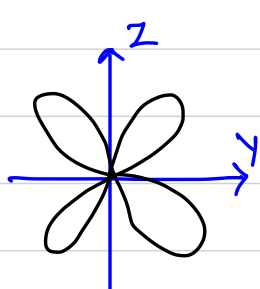
s-orbitals

along the axis orbitals

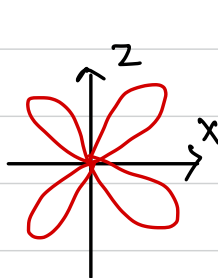
d_{xy}



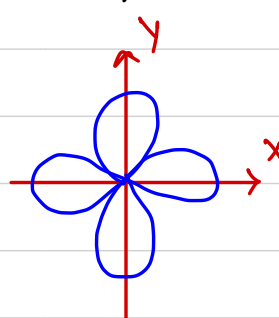
d_{yz}



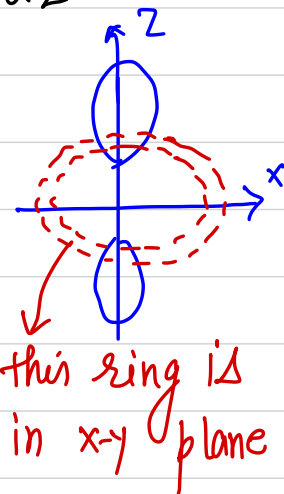
d_{zx}



$d_{x^2-y^2}$



d_{z^2}



in between the axis.

f-subshell :- no of orbitals = 7 $\rightarrow$ flower shape (complex)

s $\rightarrow$  orbital $\rightarrow$ $2e^-$

p subshell $\rightarrow$  $\times 2 = 6e^-$

d subshell $\rightarrow$  $\times 2 = 10e^-$

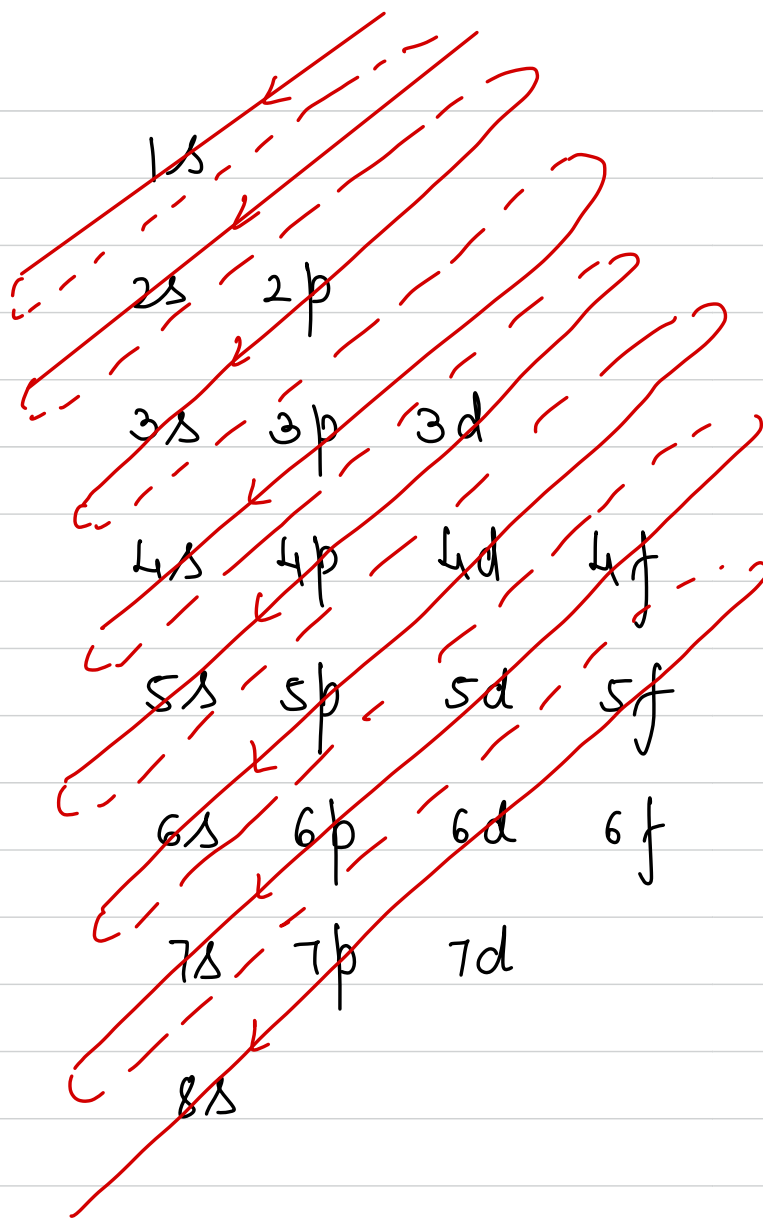
f subshell $\rightarrow$  $\times 2 = 14e^-$

Aufbau Principle :-

Auf bau
 $\downarrow$ $\downarrow$
Building up

According to this principle, the filling of e^- s in subshell occurs in increasing order of energy. i.e.

the electron will first go in lowest energy level & then to higher.



order of energy

$1s < 2s < 2p < 3s < 3p$
 $< 4s < 3d < 4p < 5s$
 $< 4d < 5p < 6s - -$
 $- - - -$

* p-subshell has 3 orbitals which are of same energy, these orbitals are called degenerate orbitals.

p-subshell has 3-fold degeneracy

d-subshell " 5-fold "

f-subshell " 7-fold "

Hund's Rule of maximum multiplicity:

According to this rule, when the filling of e^- starts in subshells, all the degenerate orbitals will first occupy one e^- with same spin. The pairing of e^- s will occur only when all the orbitals are singly occupied.

Quantum Numbers:-

The location of e^- can be set on the basis of 4 quantum numbers.

They provide us the address of the e^- in an atom.

- ① Principal Quantum no.
- ② Azimuthal Quantum no.
- ③ Magnetic Quantum no.
- ④ Spin Quantum no.

① Principal Quantum :-

shell no.
principal energy level of the electron in which it is present. It tells us about the

It is represented by 'n'.

'n' can take all positive integer values.

$n = 1, 2, 3, 4 - - -$
K L M N - - -

* If n is the no of principal energy level then n^2 will be the no of orbitals in the shell and $2n^2$ will be the no of electrons in the shell.

for $n=1$	no of orbital (n^2) 1	no of e^- s ($2n^2$) 2
$n=2$	4 2s, 2p 1+3	8
$n=3$	9	18
$n=4$	16	32

Azimuthal Quantum no:- (l)

- * It tells us about the subshell of e^- in which it is present. It is represented by ' l '.
- * Azimuthal quantum no gives us the magnitude of orbital angular momentum in z-direction.
' l ' can take values from $n=0$ to $n=n-1$.

no of l values = n

if $n=1$ then $l=0$

no of l values
1

$n=2$ then $l=0, 1$

2

$n=3$ then $l=0, 1, 2$

3

$n=4$ $l=0, 1, 2, 3$

4

$l =$	0	1	2	3
	s	p	d	f

Magnetic Quantum no:- (m)

It is represented by ' m ' or ' m_l '. It gives us the orientation of orbital in a given subshell.

* ' m ' gives us the orientation of orbital angular momentum vector

for a given value of ' l ' ' m ' can take values from $-l$ to 0 to $+l$.

$$\text{no of } m \text{ values} = 2l + 1$$

	no of m values	which value
for $l = 0$	1	0
for $l = 1$	3	$-1, 0, +1$
$l = 2$	5	$-2, -1, 0, +1, +2$
$l = 3$	7	$-3, -2, -1, 0, +1, +2, +3$

Spin Quantum no:-

It is represented by 's'. It can take two values $+\frac{1}{2}$ or $-\frac{1}{2}$.

- * It has nothing to do with clockwise and anticlockwise spin.
- * It is not the solution of schrooginder wave equation
- * It has no classical analogue.

Pauli exclusion Principle:-

No two e's in an atom can have same set of four quantum nos.

Three quantum no can be same then the fourth must be different.

Imp

$$\text{orbital angular momentum} = \sqrt{l(l+1)} \frac{h}{2\pi}$$

Imp

$$\text{spin angular momentum} = \sqrt{s(s+1)} \frac{h}{2\pi}$$

$$s = (s_1 + s_2 + s_3 + \dots)$$

Imp

$$\text{spin only magnetic moment} = \sqrt{n(n+2)} \text{ BM}$$

BM $\rightarrow$ Bohr magneton.

n is no of unpaired $e^- \rightarrow$

$$1 \text{ Bohr magneton} = 9.2732 \times 10^{-14} \text{ J}$$

Nodes:-

where the probability of finding an electron is zero.

$$\text{Number of radial nodes} = n - l - 1$$

$n \rightarrow$ principle quantum no.

$l \rightarrow$ azimuthal quantum.

$$\text{No of angular nodes or nodal planes} = l$$

$$\begin{aligned} \text{Total no of nodes} &= \text{Radial nodes} + \text{Angular nodes.} \\ &= n - l - 1 + l \end{aligned}$$

$$\text{Total no of nodes} = n - 1$$

IIT JEE 2004

Question:

For H-atom the wavefunction for 2s orbital is given by

$$\psi_{2s} = 2 \left(\frac{1}{a_0} \right)^{3/2} \left(1 - \frac{r}{2a_0} \right) e^{-r/2a_0}.$$

Then find out the position of nodes other than nucleus and infinity.

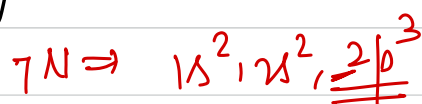
$$1 - \frac{r}{2a_0} = 0$$

$$\boxed{r = 2a_0} =$$

Calculation of quantum nos:-

0	1	2	3
s	p	d	f

Q: Calculate the set of four quantum nos for the ultimate or last e^- of N.



-1	0	+1
1	1	1

$$n=2$$

$$l=1$$

$$m=+1$$

$$s=+\frac{1}{2}$$

penultimate $e^- \rightarrow$ second last electron

(n+l) Rule:- $24Cr \Rightarrow [Ar] 4s^1 3d^5$

- * It is used to estimate the energy of a subshell. ✓
- * Greater the value of (n+l) greater will be energy.
- * If (n+l) value is same for 2 subshells then the subshell which has lower value of n will have lower energy.

